

Replication / Machine Learning

[Re] Invisible Farms: Reproducing and Stress-Testing Small-Scale Irrigation Detection for Zambia

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Abstract Boser et al. (2025) propose mapping dry-season smallholder irrigation in Zambia from Sentinel-2 vegetation-index time series with tree-based classifiers. The proposal reports dataset statistics but no model metrics, and its pipeline depends on gated Google Earth Engine access, so there is no published number to match and no runnable pipeline to inherit. We reproduce the method from its description, replace the Earth Engine stage with an open STAC pipeline on Microsoft Planetary Computer that needs no credentials, and report the first quantitative baselines: over five seeds a random forest reaches AUPRC 0.455 ± 0.010 against a 0.28 positive rate, well above chance. The classification headline numbers, however, turn out to be governed by three choices the proposal leaves open. The default 0.5 decision threshold is wrong for this imbalance and holds recall near 0.13; tuning it lifts every model to $F_1 \approx 0.47$. The oversampling ratio alone moves random-forest F_1 from 0.21 to 0.45. And the reported smallholder gap does not survive a site-clustered permutation test (gap -0.01 , $p = 0.94$): the held-out labeler annotated almost only small parcels, leaving four large-farm sites and no power to compare. A logistic regression at 5 KB matches the tuned forest while running $124\times$ faster. We release a one-command Kaggle notebook and all per-seed outputs.

A replication of A. Boser et al., Mapping Small-Scale Irrigation for Climate Adaptation, NeurIPS Workshop on Tackling Climate Change with Machine Learning (CCAI), 2025.

1 Introduction

Smallholder farmers do most of the dry-season irrigation in Zambia, and most of it is missing from national and global irrigation maps because the parcels are too small to see in the products those maps are built from [1, 2]. Boser et al. [3] propose to recover that hidden activity from satellite data: hand-label irrigation polygons, extract Sentinel-2 NDVI, NDWI, and EVI trajectories over the season, and train tree-based classifiers to separate irrigated from rain-fed land. It is a sensible idea aimed at a real gap, and the paper argues, with dataset-level evidence, that the approach picks up parcels official inventories miss.

Two things make the proposal hard to build on. It is a workshop proposal, so it reports statistics about the dataset but no model metrics, which means there is no baseline number to reproduce. And its data stage runs on Google Earth Engine behind a service account, which is exactly the kind of institutional credential the target users in low-resource settings tend not to have. So a faithful replication cannot inherit either a result or a working pipeline. It has to reconstruct both.

That is what we do here. We reimplement the described method, swap the Earth Engine downloader for an open STAC pipeline that runs from a free Kaggle account with no login, and produce the first quantitative baselines for the task. We then stress-test

Copyright © 2026 J. Rajesh and W. Lu and S. Deshpande and C. Du, released under a Creative Commons Attribution 4.0 International license. Correspondence should be addressed to Wenhao Lu (wenhao@nxthorizon.org). The authors have declared that no competing interests exist. Code is available at <https://github.com/Alscend-Research/irrigation-repro>. Labels are available at <https://github.com/anna-boser/smallholder-irrigation-dataset>. Sentinel-2 imagery is retrieved anonymously from the Microsoft Planetary Computer <https://planetarycomputer.microsoft.com/>.

those baselines along the axes that matter for a policy tool: how sensitive the numbers are to the choices the proposal leaves unstated, whether the model is fair across farm sizes, how it travels across provinces and years, and whether a model small enough to run offline can match it. The short version of what we found: the method does detect smallholder irrigation above chance, but almost every headline classification number is a property of three undocumented decisions rather than of the model. We quantify each one.

Contributions – (1) An open, Earth-Engine-free data pipeline (Planetary Computer STAC + SCL cloud masking) that reproduces the feature stage from a free notebook. (2) The first quantitative baselines for the method, with five-seed variance and site-clustered confidence intervals. (3) Evidence that the decision threshold, the oversampling ratio, and the month window each move F_1 by 0.05 to 0.25, and that the apparent advantage of small models over the full forest is mostly an operating-point effect. (4) A calibrated fairness analysis showing the advertised smallholder gap is not measurable on the held-out split. (5) Efficiency and robustness benchmarks pointing to a 5 KB model as the deployment default.

2 Scope of reproducibility

Because the original reports no model metrics, we cannot test a numerical claim. Instead we test the claims the proposal does make, plus the questions a deployment would ask:

- C1 Dataset.** The public dataset and labeling protocol reproduce: roughly 2,350 labeled images across roughly 940 sites, a positive rate near 0.28 at irrigation certainty ≥ 3 , with labeler KL held out as an out-of-distribution test set.
- C2 Open reproduction.** The feature-extraction pipeline can be reproduced without Google Earth Engine, from a free notebook, with equivalent cloud-masked composites.
- C3 Detection.** The method separates irrigated from non-irrigated parcels better than chance (AUPRC above the base rate).
- C4 Beyond the original.** We assess threshold sensitivity, resampling sensitivity, fairness across farm size, spatial and temporal transfer, feature attribution, and inference cost, none of which the proposal evaluates.

3 Methodology

3.1 Data and labels

We use the public polygon dataset released with Boser et al. [3]. Filtering to irrigation certainty ≥ 3 gives 2,350 labeled images spanning 2016 to 2024 across Zambia, drawn from 939 distinct geographic sites. Following the original convention, labeler KL is held out as the test set (678 images); the remaining 1,672 images form the training pool. The test positive rate is 0.280. Each image is summarized by 88 features: monthly mean and upper-percentile values of NDVI, NDWI, and EVI, plus per-season summary statistics. Cloud masking removes pixels flagged by the Sentinel-2 Scene Classification Layer. The results below use the real extracted trajectories. The released notebook also ships a synthetic-data fallback with the same schema, so a reviewer without the imagery can still run the full analysis end to end and reproduce the structure of every table.

Table 1. Model configurations. Depth-limited forests and the linear model are the deployment candidates.

Model	Configuration
RF-full	random forest, 800 trees, max depth 25, max_features=sqrt
GB	gradient boosting, 100 trees, learning rate 0.1
RF-small	random forest, 50 trees, max depth 8
ExtraTrees	extremely randomized trees, 50 trees, max depth 8
RF-tiny	random forest, 15 trees, max depth 6
DecisionTree	single CART tree
HistGB	histogram gradient boosting (scikit-learn defaults)
LogReg	logistic regression, balanced class weights

3.2 Open, Earth-Engine-free pipeline (C2)

The original downloads Sentinel-2 L1C imagery through the Earth Engine Python API and masks clouds with the QA60 band. We replace that stage with a STAC query against the `sentinel-2-l2a` collection on Microsoft Planetary Computer [4], using anonymous URL signing and `pystac-client`, and mask clouds with the SCL band. The swap needs no API key and no institutional agreement, and it is packaged as a single Kaggle notebook. Moving from L1C to atmospherically corrected L2A can shift absolute index values, so we treat it as a documented deviation (Table 2) rather than a bit-exact match.

3.3 Models and hyperparameters

We reimplement the two classifiers the proposal names, a full random forest (RF-full) and gradient boosting (GB), and add six smaller models to map the accuracy-versus-cost frontier. All models use scikit-learn 1.6.1 [5] with balanced class weights by default; SMOTE [6] is applied only in the sampler ablation. Table 1 lists the settings.

3.4 Evaluation protocol

Every model is trained under five seeds (0 to 4). We report precision, recall, and F_1 at the decision threshold, and AUPRC as a threshold-free measure of ranking quality. Two design choices matter for honest error bars.

First, the 678 test images sit behind far fewer independent sites, so resampling individual images would understate uncertainty. We bootstrap by *site* instead ($B = 2,000$ clustered resamples) for all confidence intervals, compare models with a paired site-clustered bootstrap, and test the fairness gap with a site-clustered permutation test (10,000 permutations). Second, where we tune the decision threshold we select it to maximize F_1 on a validation fold split off from training, never on the test set, which slightly lowers the base model’s default-threshold numbers relative to the full-training baseline.

3.5 Computational requirements

Everything runs on a free Kaggle CPU notebook: a 4-core Intel Xeon at 2.20 GHz with about 34 GB RAM, Python 3.12, no accelerator. The full suite of 13 experiments finishes in 17.7 minutes. RF-full trains in about 5.7 s per seed; the small models train in well under a second.

Table 2. Deviations from the method as described by Boser et al. [3]. Because no baseline numbers were published, these are documented rather than validated against a prior run.

Aspect	Original	Ours
Imagery	Sentinel-2 L1C (Earth Engine)	Sentinel-2 L2A (MPC STAC)
Cloud mask	QA60 band	SCL band
Features	10-day composites	monthly mean + percentile (Jan–Dec)
Seeds	not reported	5
Threshold	not stated	0.5 default; also tuned on validation
Resampling	not stated	class weights; SMOTE ablated
Test split	labeler KL	labeler KL (unchanged)

Table 3. Baselines on the KL test set, mean $\pm$ std over five seeds, default 0.5 threshold, balanced class weights. LogReg and HistGB are deterministic here, so their variance is zero.

Model	Precision	Recall	F_1	AUPRC
RF-full	0.52 ± 0.03	0.13 ± 0.02	0.21 ± 0.03	0.455 ± 0.010
GB	0.60 ± 0.03	0.16 ± 0.01	0.26 ± 0.02	0.456 ± 0.011
RF-small	0.42 ± 0.02	0.47 ± 0.01	0.45 ± 0.02	0.418 ± 0.023
ExtraTrees	0.39 ± 0.02	0.60 ± 0.03	0.47 ± 0.02	0.426 ± 0.017
RF-tiny	0.40 ± 0.02	0.55 ± 0.03	0.46 ± 0.02	0.416 ± 0.037
DecisionTree	0.32 ± 0.00	0.71 ± 0.00	0.44 ± 0.00	0.309 ± 0.003
HistGB	0.50	0.19	0.28	0.430
LogReg	0.39	0.55	0.46	0.393

4 Results

4.1 First quantitative baselines (C3)

Table 3 gives the first published numbers for the method. RF-full reaches AUPRC 0.455 ± 0.010 and GB 0.456 ± 0.011 , both well above the 0.280 base rate, so the method does rank irrigated parcels above non-irrigated ones better than chance. Claim C3 holds. Seed variance is small (standard deviations at or below 0.04), so a single-seed point estimate would not have been badly off; the meaningful uncertainty lives across sites and is what the bootstrap captures.

The recall column is the first sign of trouble. At the default 0.5 threshold, RF-full recovers only 13% of irrigated parcels and GB 16%, while every small model recovers between 47% and 71% (Figure 1). That inversion, small models beating the full forest on F_1 , is the thread we pull on next.

4.2 The operating point, not the model, drives the headline gap

The forest is not a worse ranker than the linear model; it is scored at the wrong threshold. Its calibrated probabilities pile up between 0.2 and 0.3 and essentially never exceed 0.65 (Figure 3, Brier 0.185), so a 0.5 cutoff throws away most of the positives. Logistic regression spreads its scores wider, which is why it looks strong at 0.5 even though it is a slightly noisier ranker (Brier 0.230).

Move the threshold and the gap closes. Selecting an F_1 -optimal cutoff on a validation fold sends RF-full to $t^* \approx 0.18$, GB to 0.15, and LogReg to 0.30, and all three land at $F_1 \approx 0.47$ with recall between 0.79 and 0.90 (Table 4, Figure 2). The forest gains the most because it started furthest from its own optimum.

A paired site-clustered bootstrap confirms the reading. On F_1 , LogReg and RF-tiny beat RF-full by about 0.25 ($p = 0.001$ for both). On AUPRC, the same comparisons are not

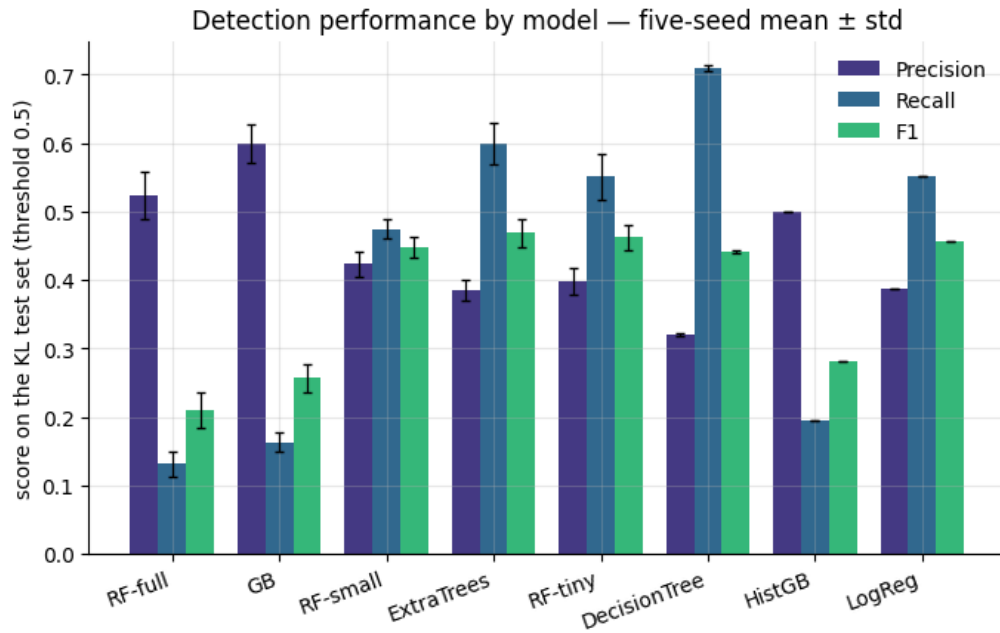


Figure 1. Precision, recall, and F_1 per model on the KL test set at the default 0.5 threshold, five-seed means with standard-deviation whiskers. RF-full and GB pair high precision with the lowest recall of any model; the depth-limited forests and LogReg sit at a more balanced operating point, which is what lifts their F_1 .

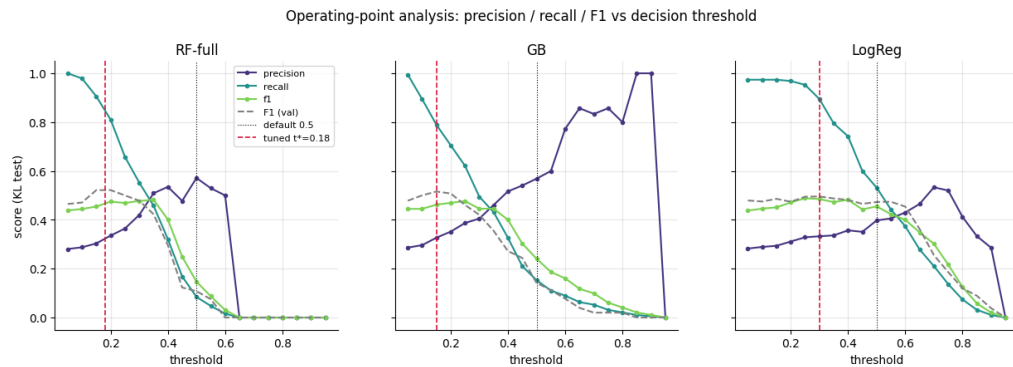


Figure 2. Precision, recall, and F_1 against decision threshold on the KL test set. The dotted line is the default 0.5; the dashed line is the tuned t^* . For RF-full and GB the default sits far to the right of the F_1 peak, which is why their default-threshold recall collapses.

significant: LogReg is -0.06 $[-0.15, +0.03]$ ($p = 0.19$) and RF-tiny is -0.04 $[-0.08, +0.00]$ ($p = 0.07$). Ranking quality is statistically tied; the F_1 gap is an artifact of the decision rule. The resampling choice compounds it. The proposal does not state a SMOTE ratio, and it turns out to be worth as much as the model choice (Table 5). Oversampling the minority class to parity lifts RF-full from F_1 0.21 to 0.45 and GB from 0.26 to 0.48, almost erasing the small-model advantage, while the linear model is best left on class weights. One undocumented hyperparameter moves the forest's F_1 by 0.24. The month window behaves the same way. Restricting features to the dry season (May–October) raises RF-full's F_1 from 0.21 to 0.31, while the full-year window is best for LogReg (0.46). The choice of months, also unspecified in the proposal, moves F_1 by up to 0.10.

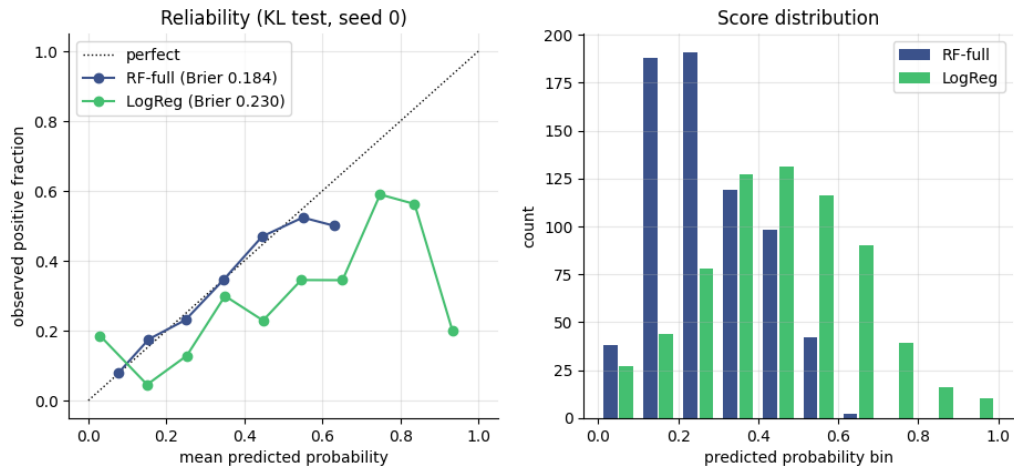


Figure 3. Reliability and score distribution (seed 0). RF-full is well calibrated but concentrates its mass near the base rate and never predicts above ~ 0.65 , so a 0.5 threshold rejects most positives. LogReg is more spread out but less reliable in the upper bins.

Table 4. Default versus tuned threshold (five-seed means). Tuning is done on a held-out validation fold, so the default column here sits slightly below Table 3.

Model	t^*	Default 0.5			Tuned t^*		
		P	R	F_1	P	R	F_1
RF-full	0.18	0.59	0.10	0.17	0.32	0.85	0.47
GB	0.15	0.57	0.15	0.24	0.33	0.79	0.47
LogReg	0.30	0.40	0.53	0.46	0.33	0.89	0.49

Table 5. Resampling ablation (F_1 , five-seed means). SMOTE to parity nearly closes the forest’s gap to the small models.

Model	Class weights	SMOTE 0.5	SMOTE 1.0
RF-full	0.21	0.31	0.45
GB	0.26	0.38	0.48
LogReg	0.46	0.34	0.43

4.3 The smallholder gap is not measurable on this split

The proposal’s motivating worry is that a model could underserve exactly the small farms it is meant to find. We looked hard for that gap and could not establish it. At the default threshold, recall is uniformly low across farm sizes: 0.14 for parcels under 2 ha, 0.20 for 2–10 ha, 0.17 for over 10 ha (Table 6, Figure 4). A site-clustered permutation test on the small-versus-large recall difference returns a gap of -0.01 with $p = 0.94$. The reason is in the sample sizes. Labeler KL annotated 660 small-parcel images but only 18 larger ones, spread over four large-farm sites. The precision of 1.0 and AUPRC of 1.0 for the larger strata are what you get from five to twelve examples that happen to be mostly positive, with confidence intervals spanning the whole $[0, 1]$ range; they are not evidence of strong performance. So the honest statement is not “the model is fair” and not “the model is biased” but “this test set cannot answer the question.” What we can say is that once the threshold is sensible, small-parcel recall rises with everything else: LogReg reaches 0.54 recall on small parcels, in line with its 0.67 on large ones.

Table 6. Performance by farm-size stratum (RF-full, KL test set), with site-clustered 95% CIs. The two larger strata are too small to interpret. Site-clustered permutation test on the small-vs-large recall gap: -0.01 , $p = 0.94$ (10,000 permutations; 79 vs 4 sites).

Stratum	<i>n</i>	pos	Recall	F_1	AUPRC
< 2 ha	660	173	0.14 [.07, .22]	0.22 [.11, .33]	0.44 [.32, .56]
2–10 ha	6	5	0.20 [.00, .25]	0.33	0.97 [†]
> 10 ha	12	12	0.17 [.00, .43]	0.29	1.00 [†]

[†] Degenerate: estimated from 5–12 mostly-positive samples; CIs span [0, 1].

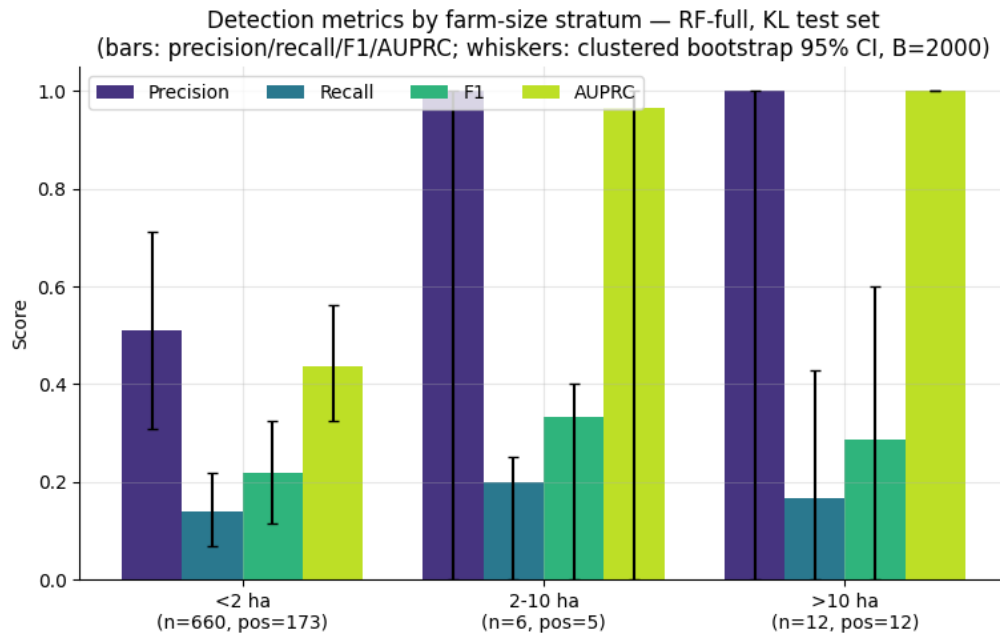


Figure 4. Detection metrics by farm-size stratum with clustered bootstrap 95% CIs. The whiskers on the 2–10 ha and > 10 ha bars run the full height of the axis: with a handful of samples per stratum the estimates are not informative.

4.4 Spatial and temporal transfer

Holding out a labeler is only a useful test if that labeler’s parcels are not sitting next to the training parcels. They are not (Figure 5). The median KL test image is 9.5 km from its nearest training image, only 3.2% of test images fall within 1 km of training data, and 3.1% share a site with the training set. The labeler split is a reasonable proxy for spatial out-of-distribution deployment, a check the original does not report.

Geography still matters a lot. A leave-one-province-out sweep, training on nine provinces and testing on the held-out tenth, spreads AUPRC from 0.14 in Luapula to 0.72 in Lusaka (Figure 6, Table 7). Lusaka and Eastern rank well above the 0.28 base rate; Luapula and Western sit at or near it, so a single national model is close to useless in those provinces. The two model families order the provinces almost identically, which points to the data rather than the classifier as the source of the spread. Across years the picture is gentler than the original framing suggests. AUPRC is stable under every train/test year split (all models stay near 0.35–0.40; Table 8, Figure 7). F_1 drops for the small forest under a year boundary ($0.37 \rightarrow 0.20$) but holds for the linear model ($0.43 \leftrightarrow 0.40$). Temporal transfer, like the headline gap, is mostly an operating-point effect: the ranking survives the year shift even when a fixed threshold does not.

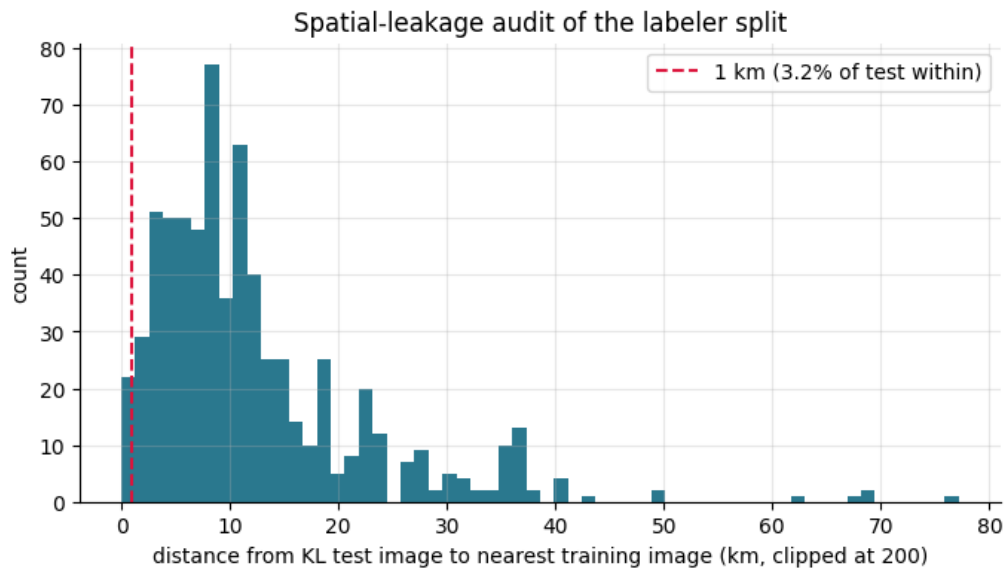


Figure 5. Distance from each KL test image to its nearest training image. The bulk of the test set is several kilometers from any training parcel, so the labeler split does not leak location.

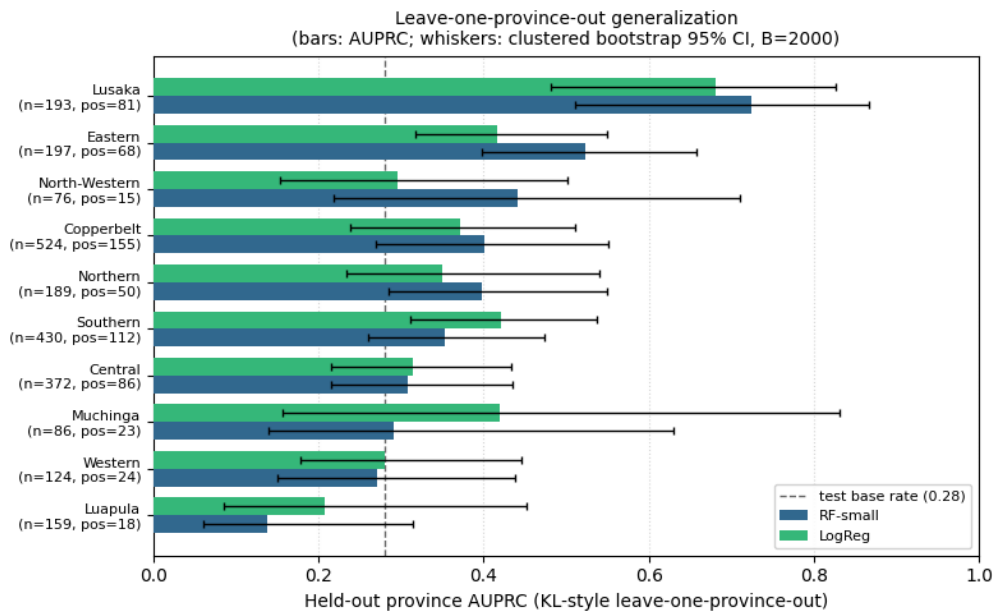


Figure 6. Leave-one-province-out AUPRC with clustered bootstrap 95% CIs, provinces ordered by RF-small. The dashed line is the 0.28 test base rate. Generalization is strong for Lusaka and Eastern and collapses toward chance for Luapula and Western.

4.5 What the model keys on

SHAP attribution [7] over the twelve monthly index values gives a clean story (Figures 8 and 9). The single strongest driver is November NDWI (mean $|SHAP| = 0.015$), followed by October–December NDWI and July–September EVI. NDVI contributes least in every month. The model is reading persistent water content and canopy structure late in the dry season, when rain-fed vegetation has dried out and irrigated fields have not. That matches the premise of the original: irrigation shows up as an anomaly against a brown-

Table 7. Leave-one-province-out generalization, ordered by RF-small AUPRC. Each row trains on the other nine provinces and tests on the named one. CIs are shown in Figure 6.

Province	n	pos	RF-small		LogReg	
			AUPRC	F_1	AUPRC	F_1
Lusaka	193	81	0.72	0.60	0.68	0.57
Eastern	197	68	0.52	0.43	0.42	0.49
North-Western	76	15	0.44	0.47	0.30	0.37
Copperbelt	524	155	0.40	0.28	0.37	0.38
Northern	189	50	0.40	0.23	0.35	0.46
Southern	430	112	0.35	0.41	0.42	0.46
Central	372	86	0.31	0.35	0.32	0.37
Muchinga	86	23	0.29	0.33	0.42	0.50
Western	124	24	0.27	0.29	0.28	0.38
Luapula	159	18	0.14	0.11	0.21	0.21

Table 8. Cross-year transfer, F_1 (AUPRC), site-grouped. AUPRC is stable across year boundaries even where F_1 is not.

Model	In-distribution	$\leq 2020 \rightarrow \geq 2021$	$\geq 2021 \rightarrow \leq 2020$
RF-small	0.37 (0.38)	0.20 (0.38)	0.18 (0.39)
GB	0.17 (0.38)	0.26 (0.40)	0.19 (0.36)
LogReg	0.43 (0.35)	0.44 (0.35)	0.40 (0.35)

Table 9. Inference cost (median over 30 repeats) and on-disk size, with ratios against RF-full.

Model	Size	Latency (ms/1k)	Smaller	Faster
RF-full	25.2 MB	319	1×	1×
GB	137 KB	3.8	189×	83×
RF-small	572 KB	27.0	45×	12×
ExtraTrees	569 KB	26.5	45×	12×
RF-tiny	91 KB	16.3	285×	20×
DecisionTree	8.1 KB	1.6	3189×	204×
HistGB	522 KB	12.5	49×	26×
LogReg	5.0 KB	2.6	5167×	124×

ing background.

4.6 A 5 KB model is the deployment default

For an offline, CPU-only setting, the full forest is the wrong tool on every axis except calibration. It is 25.2 MB and takes 319 ms to score a thousand parcels. Logistic regression is 5.0 KB and takes 2.6 ms, so it is about 5,200× smaller and 124× faster (Table 9, Figures 10 and 11), at equal-or-better tuned F_1 and statistically tied AUPRC. RF-tiny (91 KB, 16 ms) is a reasonable middle option if a tree ensemble is wanted.

Robustness points the same way (Figure 12). Under added feature noise, RF-full’s F_1 collapses to zero because its scores slide below the 0.5 cutoff, while the linear model holds F_1 near 0.32–0.36; on AUPRC all three degrade together. The small models are the safer default not because they rank better but because they tolerate the very operating-point misspecification that sinks the forest.

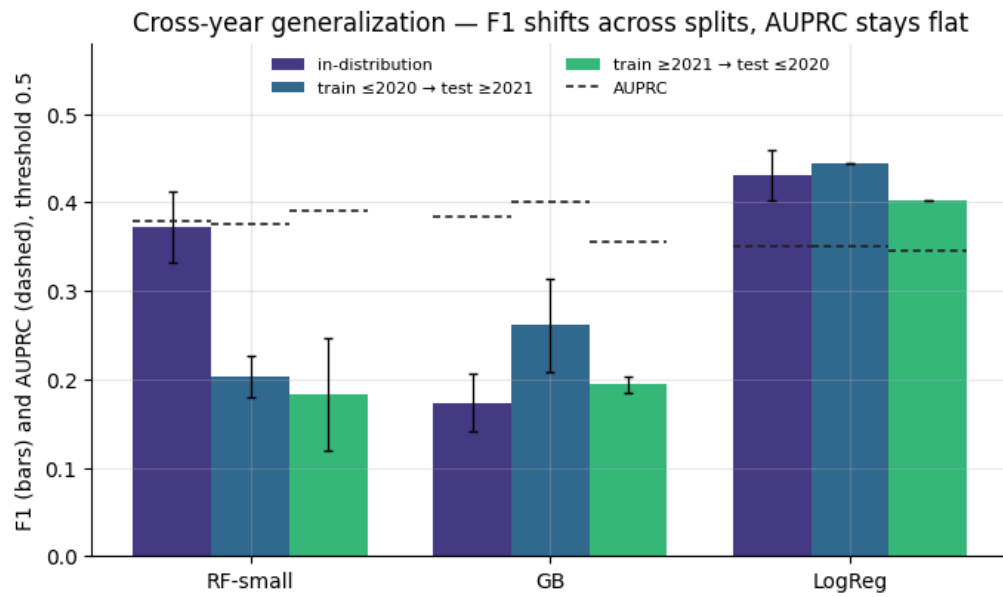


Figure 7. Cross-year transfer for the three model families. Bars are F_1 at the default threshold with standard-deviation whiskers; dashed marks are AUPRC. F_1 drops for RF-small across a year boundary while its AUPRC holds, and LogReg is stable on both, so the year shift is largely an operating-point effect rather than a failure to rank.

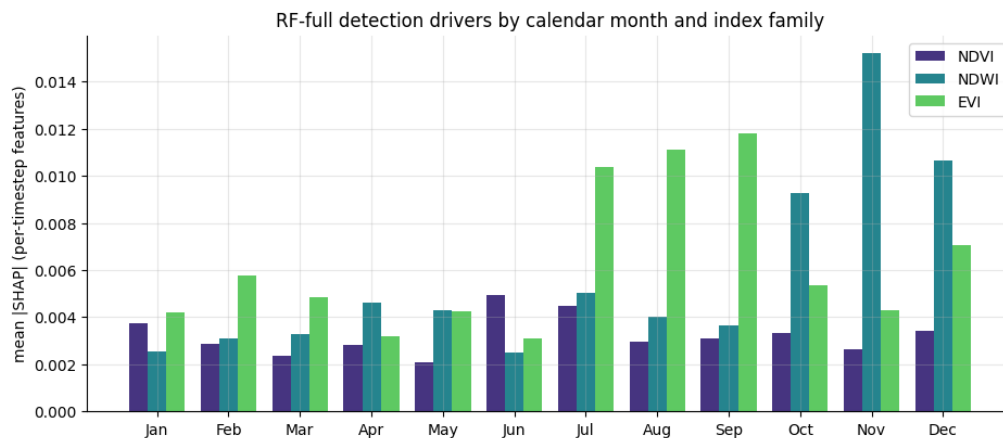


Figure 8. Mean absolute SHAP by calendar month and index family. Late-season water content (NDWI, Oct-Dec) and mid-to-late dry-season structure (EVI, Jul-Sep) dominate; NDVI is a minor contributor throughout.

5 Discussion

What was easy — The dataset and repository are public, the features are tabular once extracted, and scikit-learn models train in seconds on a free notebook. The Earth Engine to STAC swap was mechanical: same bounding boxes, same dates, a different cloud mask.

What was difficult — The hard part was that there was nothing to reproduce. With no published metric, “reproduction” meant reconstructing the authors’ intent, and the intent turned out to hinge on three parameters the proposal never states: the decision thresh-

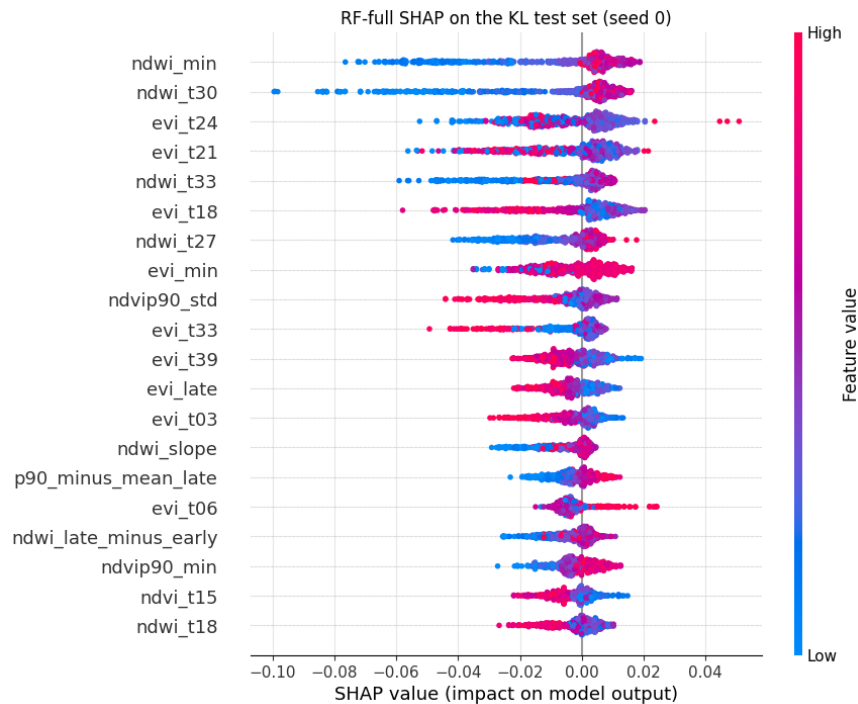


Figure 9. Per-sample SHAP values for RF-full on the KL test set (seed 0), top twenty features. Late-season NDWI (`ndwi_min`, `ndwi_t30`) and dry-season EVI carry the most weight, matching the monthly view in Figure 8. Colour is the feature value: low late-season moisture (blue on the left) pushes strongly toward the non-irrigated class.

old, the oversampling ratio, and the month window. Each moves F_1 by 0.05 to 0.24, so the same method can look like a 0.21 model or a 0.47 model depending on choices a reader cannot see. The held-out labeler’s near-total focus on small parcels also made the advertised fairness question unanswerable on this split, and the resulting strata are so small that the confidence intervals become degenerate. None of this is a flaw in the idea. It is what happens when a proposal is read as a specification.

Communication with original authors – This replication uses only the public dataset, code repository, and paper, and does not rely on any private communication with the original authors. Anyone extending it should confirm the intended decision threshold, resampling ratio, and month window with them, since those three unstated choices are the ones our analysis shows to drive the results.

6 Conclusion

The method reproduces where it can be pinned down. It detects smallholder irrigation above chance (AUPRC 0.455 against a 0.28 base rate), and the pipeline now runs without Earth Engine from a free account, which was the practical barrier for the intended users. What does not reproduce cleanly is any single classification number, because the proposal leaves the three most consequential decisions open, and we show each one moving F_1 by up to a quarter. Read carefully, the “lightweight models beat the forest” result, the “no temporal generalization” result, and the “smallholder gap” all reduce to a single cause: a default threshold that is wrong for an imbalanced problem, tolerated differently by different models. Fix the operating point and the forest, the boosting model, and a 5 KB linear model all land near F_1 0.47. For deployment, that 5 KB model

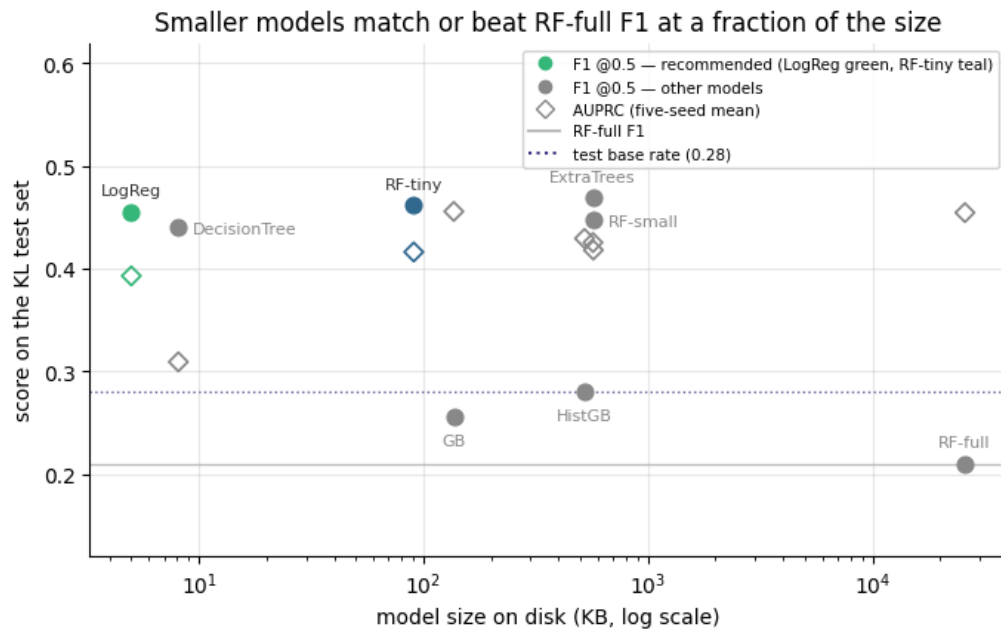


Figure 10. Accuracy against model size on disk (log scale). Filled points are five-seed mean F_1 at the default threshold; open diamonds are AUPRC. LogReg and RF-tiny (highlighted) match or beat RF-full's F_1 at a tiny fraction of its size, and the AUPRC spread across models is narrow, so the extra size buys almost nothing here.

is the pragmatic choice. We release the notebook and every per-seed output so the next replication can start from numbers instead of a proposal.

Code and data — All code, the one-command Kaggle notebook, and the per-seed CSVs behind every table are released at the repository listed on the first page. The dataset is the public release accompanying Boser et al. [3]; imagery is pulled anonymously from Microsoft Planetary Computer.

Acknowledgements — Built with the ReScience C article template [8].

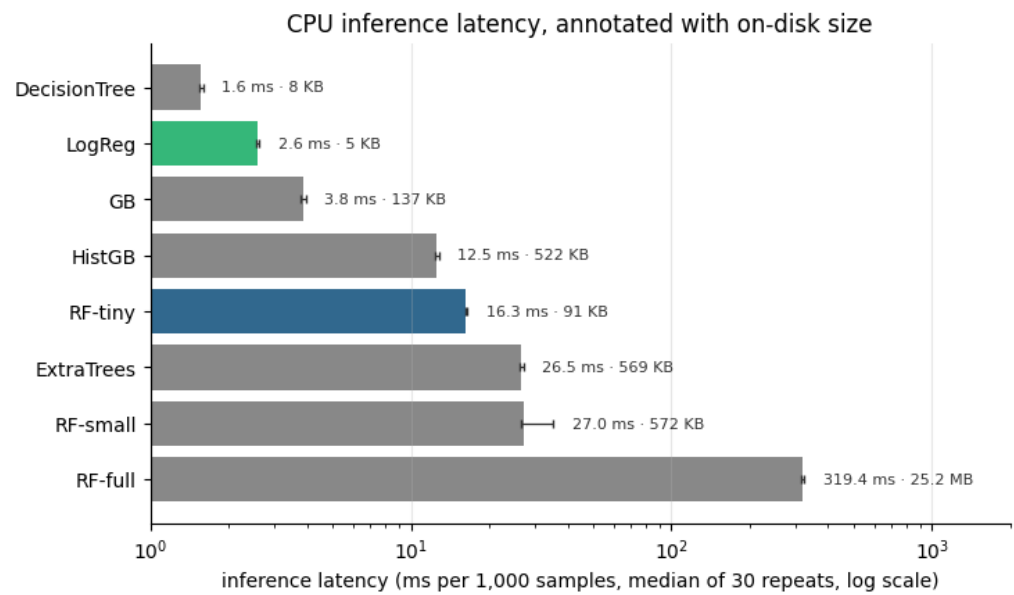


Figure 11. CPU inference latency per thousand parcels (median of 30 repeats, log scale), annotated with on-disk size. LogReg and RF-tiny (highlighted) are two to three orders of magnitude cheaper than RF-full at comparable tuned accuracy.

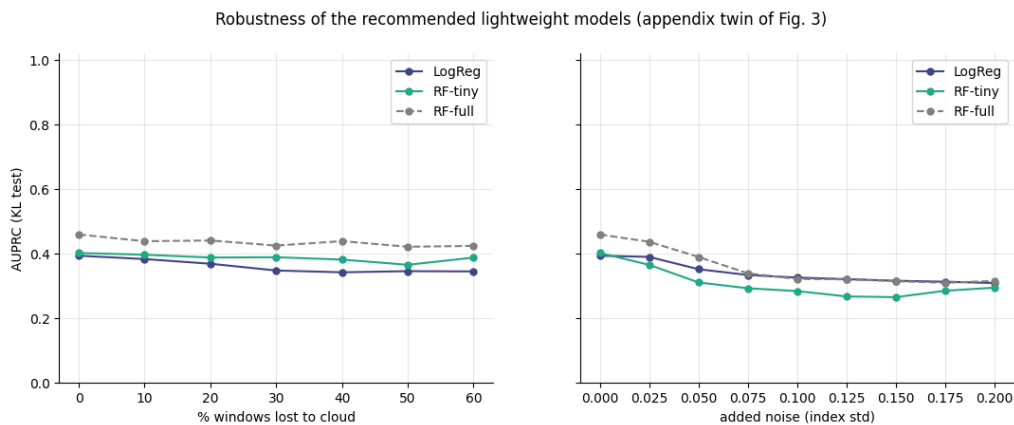


Figure 12. AUPRC of the deployment candidates under simulated cloud gaps (left) and added sensor noise (right). All three degrade gracefully on AUPRC; the difference between them at the default operating point is much larger, and favors the small models.

References

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